

PATHAHEAD · FREE ROADMAP

Become an AR/VR (XR) Developer from Scratch

Build worlds people can step into. Go from zero to shipping AR apps on phones and VR experiences on headsets in about five months — the skills behind gaming, training, and the spatial web.

22

weeks · ~10–12 hrs/wk

5

XR projects

4

phases · to published XR

₹0

free tools · phone AR

A free PathAhead learning resource · adjust the pace to your life — the order matters more than the dates.

THE ONE RULE

Prototype in 3D → test on device → optimize

XR only feels real on a device. Build small, run it on your phone (or headset) early, and optimize relentlessly — a smooth simple scene beats a beautiful slideshow every time.

Legal & ethical use only. No headset needed to start: a recent Android phone covers all of mobile AR, and Unity's XR Device Simulator lets you build VR interactions before you ever touch a Quest.

THE ROADMAP

Four phases, step by step

0

Foundations

Weeks 1–5

C#, Unity, and thinking in three dimensions.

WEEK 1

C# basics

Variables, methods, classes — Unity's language.

WEEK 2

Unity editor

Scenes, GameObjects, components, play mode.

WEEK 3

3D intuition

Vectors, transforms, rotation — without heavy math.

WEEK 4

Physics & interaction

Rigidbody, colliders, triggers, input.

WEEK 5

Git for Unity

Version control that survives binary assets.

1

Unity Depth

Weeks 6–11

From empty scene to polished mobile AR.

WEEK 6

UI & prefabs

Reusable objects and world-space interfaces.

WEEK 7

Animation & audio

Animator, timelines, spatial sound basics.

WEEK 8

Materials & lighting

Make scenes look good on a budget.

WEEK 9

AR Foundation

Plane detection and anchors on your own phone.

WEEK 10

AR interaction

Place, scale, and manipulate virtual objects.

WEEK 11 · PROJECT ►

AR placement app

A furniture/poster previewer running on your phone.

2

VR Development

Weeks 12–17

Presence, hands, and comfort — real VR skills.

WEEK 12

XR Interaction Toolkit

Rigs, controllers, grabbing and UI in VR.

WEEK 14

Hands & interaction design

Natural grabbing, poking, and feedback.

WEEK 16

Polish & haptics

Sound, vibration, and juice that sells presence.

WEEK 13

Locomotion

Teleport vs smooth movement — comfort first.

WEEK 15

Standalone optimization

Hit frame rate on Quest-class hardware.

WEEK 17 · PROJECT ►

VR experience

A room-scale game or training scene (simulator OK).

3

Ship & Portfolio

Weeks 18–22

XR UX, performance, and published proof.

WEEK 18

XR UX rules

Comfort, motion sickness, readability in space.

WEEK 20

WebXR taste

A browser-based XR scene anyone can open.

WEEK 22 · CAPSTONE ►

Published XR app

A polished AR or VR project, public and documented.

WEEK 19

Profiling

Find and fix what's eating your frame rate.

WEEK 21

Publishing

Play Store for AR; App Lab route for VR.

BUILD A PORTFOLIO

Five projects that get you hired

Projects beat certificates. Do them in order — each builds on the last. Put every one on GitHub with a clear README: problem → approach → result.

Start here

1 • 3D Mini-Game

A simple Unity game — proves C#, physics, and scene skills.

Mobile AR

2 • AR Placement App

Place and scale 3D furniture in your real room via phone.

VR

3 • VR Interaction Scene

Grab, throw, and manipulate objects with full comfort options.

In demand

4 • AR Learning App

An educational AR experience — anatomy, planets, or machines.

Capstone

5 • Published XR Experience

A complete app on the Play Store or shared as WebXR.

ROLES YOU CAN APPLY FOR

AR/VR Developer · Unity Developer · XR Interaction Designer · 3D Application Developer

TYPICAL STARTING SALARY · INDIA

₹4–12 LPA

LEARN IT

Learning resources

Every item links to the real source (clickable in this PDF).

BOOKS

- [Augmented Reality Principles and Practice](#) by Dieter Schmalstieg and Tobias Hollerer
- [Learning Virtual Reality](#) by Tony Parisi
- [Understanding Virtual Reality](#) by William Sherman and Alan Craig

COURSES

- [Unity Learn XR](#) (free)
- [Introduction to Virtual Reality](#) by Coursera
- [Introduction to Augmented Reality and ARCore](#) by Coursera
- [XR Development](#) by Meta on Coursera

YOUTUBE

- [Valem Tutorials](#)
- [Dilmer Valecillos](#)
- [Brackeys](#)

PRACTICE

- [Unity Asset Store](#)
- [GitHub](#)
- [Meta Quest Developer Hub](#)

DO THIS TODAY

Your first step (takes 30 minutes)

1

Install Unity Hub

unity.com — free personal license, the industry-standard XR engine.

2

Do 'Create with Code' hour one

Unity Learn's free path — you'll move a player around today.

3

Run the AR sample on your phone

AR Foundation samples — see a cube sit on your real desk.

Job-ready checklist: 3–5 solid projects on GitHub (≥ 1 deployed or documented) · can explain your work in an interview · comfortable with the core tools of the field · a portfolio or profile recruiters can find.

PathAhead — India's premium career guidance platform. This roadmap is a free educational resource; timelines are a guide and everyone learns at a different pace. Referenced third-party resources (Unity Learn, Meta Quest Developer Hub, Google Codelabs, Valem Tutorials, Brackeys) are for learning only and are not affiliated with PathAhead.